Q1. What is a parameter?

Ans1. A **parameter** in machine learning is a value that the model learns from the training data. These values help the model make predictions. For example, in a linear model like y=mx+by = mx + b, **m** and **b** are parameters.

Q2. What is correlation? What does negative correlation mean?

Ans2. Correlation is a measure of how two things are related. If one changes, does the other change too? It tells us if there's a connection between them.
Negative correlation means that when one thing goes up, the other goes down. For example, if more people exercise, their weight might go down — that's a negative correlation.

Q3. Define Machine Learning. What are the main components in Machine Learning?

Ans3. Machine Learning is a way for computers to learn from data and make decisions or predictions without being explicitly programmed. It's like teaching a computer to recognize patterns and improve over time.

Main components of Machine learning are:
Data: The raw information used to train the model (e.g., images, numbers, text).
Model: The system that learns from data and makes predictions (e.g., decision tree, neural network).
Algorithm: The method used to train the model (e.g., linear regression, k-nearest neighbors).
Training: The process of feeding data to the model so it can learn patterns.
Evaluation: Testing how well the model performs using new, unseen data.
Prediction: Using the trained model to make decisions or forecasts.

Q4.How does loss value help in determining whether the model is good or not?

Ans4. Loss value tells us how far off the model's predictions are from the actual answers. It's like a score for how wrong the model is.

- **Low loss** means the model is doing well — its predictions are close to the correct answers.
- **High loss** means the model is struggling — its predictions are far from the truth.

Think of it like a test: the fewer mistakes, the better the score. So, a good model usually has a **low loss value**

Q5. What are continuous and categorical variables?

Ans5. Continuous Variables: These are numbers that can take any value within a range. You can measure them, and they often include decimals.

- Examples: height, weight, temperature, income

Categorical Variables: These are values that represent groups or categories. They describe qualities, not quantities.

- Examples: gender, color, type of car, country

Q6. How do we handle categorical variables in Machine Learning? What are the common techniques?

Ans6.Categorical variables need to be converted into numbers so machine learning models can understand them. This process is called **encoding**.

Label Encoding

- Each category is assigned a unique number.
- Example: `Red = 0`, `Blue = 1`, `Green = 2`

One-Hot Encoding

- Creates a new column for each category with 0s and 1s.
- Example: `Red = [1, 0, 0]`, `Blue = [0, 1, 0]`, `Green = [0, 0, 1]`

Ordinal Encoding

- Used when categories have a natural order.
- Example: `Low = 1`, `Medium = 2`, `High = 3`

Target Encoding

- Replaces categories with the average value of the target variable for each category.
- Useful in some models but can lead to overfitting.

Q7. What do you mean by training and testing a dataset?

Ans7. Training means teaching the machine learning model using known data. The model looks at the input and the correct output to learn patterns.
Example: You show the model pictures of cats and dogs with labels, so it learns to tell them apart.

Testing means checking how well the model performs on new, unseen data. This helps us know if the model can make good predictions in real life.
Example: You give the model new pictures (without labels) to see if it correctly guesses “cat” or “dog.”

Q8. What is sklearn.preprocessing?

Ans8. sklearn.preprocessing is a module in **Scikit-learn**, a popular Python library for machine learning. This module helps you prepare your data before feeding it into a model.

It includes tools to:

- **Scale** numbers (e.g., make all features between 0 and 1)
- **Normalize** data (e.g., adjust values to have similar ranges)
- **Encode** categorical variables (e.g., convert text labels to numbers)
- **Handle missing values** (e.g., fill in blanks with averages)

Q9. What is a Test set?

Ans9. A **test set** is a portion of your data used to check how well your machine learning model performs on **new, unseen data**.

- It helps you see if the model can make accurate predictions in the real world.
- It's like giving the model an exam after it has studied (trained) on other data.

The test set is **not used during training —** it's only for evaluation.

Q10. How do we split data for model fitting (training and testing) in Python? How do you approach a Machine Learning problem?

Ans10.

Q11. Why do we have to perform EDA before fitting a model to the data?

Ans11. EDA (Exploratory Data Analysis) helps us understand the data before building a model. It's important because:

1. **Detects Missing Values** – So we can handle them properly.
2. **Finds Outliers** – Which may affect model performance.
3. **Reveals Data Distribution** – Helps choose the right algorithms.
4. **Shows Relationships** – Between features and target variable.
5. **Identifies Data Errors** – Like duplicates or wrong formats.
6. **Guides Feature Selection**– We learn which features are useful.

Q12. What is correlation?

Ans12. **Correlation** is a statistical measure that shows how two variables move in relation to each other.

- If both variables increase together → **positive correlation**
- If one increases while the other decreases → **negative correlation**
- If there's no consistent pattern → **no correlation**

The most common measure is the **Pearson correlation coefficient (r)**:

- Ranges from **-1 to +1**
  - **+1** = perfect positive correlation
  - **-1** = perfect negative correlation
  - **0** = no correlation

Example: Height and weight usually have a **positive correlation**—taller people tend to weigh more.

Q13. What does negative correlation mean?

Ans13. **Negative correlation** means that **as one variable increases, the other decreases**.

- Example: If study time goes up and number of mistakes goes down, they have a negative correlation.
- The correlation coefficient (r) is **less than 0**, and can go down to **-1** for a perfect negative relationship.

Think of it like a seesaw—when one side goes up, the other goes down.

Q14.How can you find correlation between variables in Python?

Ans14. In Python, correlation is commonly calculated using the **Pandas** library:

```
import pandas as pd
df = pd.read_csv('data.csv')
correlation_matrix = df.corr()
```

- df.corr() returns the correlation between all numeric columns.
- It uses **Pearson correlation** by default.

Q15. What is causation? Explain difference between correlation and causation with an example.

Ans15. **Causation** means that **one variable directly affects another**. It shows a **cause-and-effect** relationship.

- Example: If you increase the temperature of water, it **causes** it to boil. Temperature is the cause, boiling is the effect.

Aspect	Correlation	Causation
Meaning	Variables move together	One variable causes change in another
Direction	Can be positive, negative, or none	Always has a cause → effect direction

Proof Needed	Simple statistical measure	Requires deeper analysis or experiments
Example	Ice cream sales increased and drowning cases increased	Smoking causes lung cancer

Q16. What is an Optimizer? What are different types of optimizers? Explain each with an example.

Ans16. An **optimizer** is an algorithm used in machine learning to **adjust the model's parameters (like weights)** to minimize the error (loss function). It helps the model learn by improving its predictions step by step.

Optimizer	Key Feature	Example Use Case
Gradient Descent	Full dataset per update	Linear regression
SGD	One sample per update	Deep learning
Mini-Batch	Small batch per update	Neural networks
Momentum	Uses past gradients	Faster convergence
RMSprop	Adaptive learning rate	RNNs
Adam	Combines Momentum + RMSprop	CNNs, NLP models

Q17. What is sklearn.linear_model ?

Ans17. sklearn.linear_model is a module in the scikit-learn library that provides linear models for machine learning tasks like regression and classification.
It includes popular algorithms that assume a linear relationship between input features and the output.

Model	Purpose	Example Use Case
`LinearRegression`	Regression	Predicting house prices

`LogisticRegression`	Classification	Spam detection (yes/no)
`Ridge`	Regularized regression	Prevents overfitting with L2 penalty
`Lasso`	Feature selection	Shrinks less important features
`SGDClassifier`	Fast classification	Large-scale text classification

Q18.What does model.fit() do? What arguments must be given?

Ans18. model.fit() is a method used to **train a machine learning model**. It takes the input data and learns the relationship between features and target values.

- It adjusts the model's internal parameters (like weights) to **minimize error**.
- After fitting, the model is ready to make predictions using model.predict()

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train, y_train)
```

Q19. What does model.predict() do? What arguments must be given?

Ans19. model.predict() is used to **make predictions** using a trained machine learning model. It takes new input data and returns the model's predicted output.

- It uses the parameters learned during model.fit() to generate results.

```
model.predict(X)
predictions = model.predict(X_test)
```

Q20. What are continuous and categorical variables?

Ans20. Repeat question same as Question 5

Q21. What is feature scaling? How does it help in Machine Learning?

Ans20. Feature scaling is a technique used to **normalize or standardize** the range of independent variables (features) in a dataset.

- It ensures that all features contribute equally to the model.
- Especially important when features have different units or scales (e.g., age vs income).

It helps in Machine learning by:

- Improves Model Accuracy
- Speeds Up Convergence
- Prevents Bias Toward Larger Values
- Essential for Distance-Based Models

Q22.How do we perform scaling in Python?

Ans22. Scaling is the process of transforming numerical features so they fall within a similar range. This is important because many machine learning algorithms perform better when features are on the same scale.
In Python, scaling is typically done using the sklearn.preprocessing module from the scikit-learn library.

Q23. What is sklearn.preprocessing?

Ans23. sklearn.preprocessing is a module in the scikit-learn library that provides tools to prepare and transform data before feeding it into a machine learning model.

It helps improve model performance by making sure the data is clean, scaled, and formatted correctly.

Q24. How do we split data for model fitting (training and testing) in Python?

Ans24. We use the **train_test_split()** function from the **scikit-learn** library to divide data into training and testing sets.

Syntax:

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

This helps the model learn from one part of the data (training) and get evaluated on unseen data (testing) to check its performance.

Q25. Explain data encoding?

Ans25. Data encoding is the process of converting categorical data (like names, labels, or categories) into numerical format so that machine learning models can understand and use it.
Most ML algorithms work with numbers—not text—so encoding is essential for handling non-numeric features.